

TED UNIVERSITY

SENG 491-SENIOR PROJECT I

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Berre Yazgı 41255097216

Tunahan Balcı 12865148094

Supervisor:
Mehmet Evren ÇOŞKUN

Jury Members:
Ayşenur BİRTÜRK
Eren ULU

Project Name: Dizoda Furniture Engine

Project URL: https://dizoda-furniture-engine.pages.dev/

Project Description:

Dizoda Furniture Engine is the furniture placement engine subsystem of Dizoda, a mobile application that helps users furnish a real room with real products. Using the measured room model produced by the app's AR room scan, which includes the walls, ceiling height, doors, windows, furniture that stays and the position from which the room photograph was taken, the engine decides exactly where each chosen product goes and which way it faces. Most current AI interior design tools let a generative model place furniture directly in the picture, which often leads to pieces that overlap, block doors or simply do not fit in the real room. This project approaches furniture placement as a constrained optimization problem instead. Strict rules, such as keeping every piece inside the room, avoiding collisions, keeping doors and windows clear and leaving enough walking space, must be satisfied by every layout the engine returns. Softer design preferences, such as placing related pieces together, facing furniture in sensible directions and keeping the room balanced and visible in the photo, are used to rank the results. The engine offers several distinct alternative layouts and, when a product cannot be placed, explains why and suggests the largest size that would fit.

Because the engine knows where the photograph was taken, it can also calculate where each placed piece appears in the room photo, so the furnished image and the plan always match. The application will show each layout on a scaled, top-down room plan with numbered items that correspond to the photo. Users will be able to choose between alternative layouts, swap individual pieces for alternative products that fit, and arrange the furniture themselves by moving and rotating it, receiving instant feedback on whether each change is still valid, along with an explanation of any broken rule. The engine works on its own with any set of products of known size, and it can also be integrated with the AI subsystem of the app, acting as an adapter that turns the products chosen by the AI into furniture placements that fit the real room. It will be evaluated on a set of real rooms, aiming for layouts that never break the strict rules, are generated within seconds, and are preferred by users in blind comparisons.